



10790 - LAPIS: Large Area Parallel Imaging Survey

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
Tier-1				
	1		NIRCam Imaging	PARALLEL
	2		NIRCam Imaging	PARALLEL
Tier-2				
	3		NIRCam Imaging	PARALLEL
	4		NIRCam Imaging	PARALLEL
	5		NIRCam Imaging	PARALLEL
Tier-3				
	6		NIRCam Imaging	PARALLEL
	7		NIRCam Imaging	PARALLEL
	8		NIRCam Imaging	PARALLEL
	9		NIRCam Imaging	PARALLEL
Tier-4				
	10		NIRCam Imaging	PARALLEL
	11		NIRCam Imaging	PARALLEL
	12		NIRCam Imaging	PARALLEL
	13		NIRCam Imaging	PARALLEL

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	14		NIRCam Imaging	PARALLEL

ABSTRACT

The abundance of UV-bright galaxies and the properties of spectroscopically confirmed systems at $z=10-14$ reveal a surprisingly mature Universe only 300 Myr after the Big Bang. These discoveries motivate the Large Area Parallel Imaging Survey (LAPIS), a pure-parallel program designed to uncover the first generation of galaxies at $z>16$, within the first 250 Myr of cosmic history. LAPIS is optimized to operate efficiently during single mechanical-move opportunities - the most common and previously underused parallel configurations - while employing a smart tiering system that flexibly adds filters when additional mechanism moves are available.

With over 100 independent sightlines, LAPIS will cover over 1000 arcmin², sampling a cosmic volume of $\sim 7 \times 10^6$ cMpc³ ($z=16-25$) and targeting UV-bright galaxies at the unexplored redshift frontier. The survey will constrain the UV luminosity density, dust content, and morphology of the earliest galaxies, providing the first robust census of luminous systems at $z>16$. Depending on the underlying physics of early star formation, LAPIS is expected to detect between a few and several dozen such galaxies, enough to distinguish between feedback-free starburst and stochastic formation models. Beyond its primary goal, LAPIS will deliver extensive legacy science, identifying rare quiescent galaxies, dusty starbursts, and Little Red Dots at intermediate redshifts. All reduced data, value-added catalogs, and completeness tools will be released promptly, ensuring broad community impact for years to come.

OBSERVING DESCRIPTION

— A High-Efficiency Pure-Parallel NIRCam Imaging Program —

Overview.

We will use NIRCam imaging in pure-parallel mode to identify bright $z>16$ candidates over previously unexplored area. Because pure-parallel opportunities are constrained by the number of available mechanism moves in the primary observation, LAPIS is optimized to deliver its core science with a single move, while scaling—via a smart tiering scheme—when more moves are available. All pointings are placed at high Galactic latitude ($|b| > 25^\circ$) to minimize stellar contamination. With simultaneous SW+LW imaging, the Tier I configuration executes a deep dropout plus two red continuum bands, yielding robust $z>16$ selections suitable for NIRSpec follow-up (<10 hr).

Exposure setup.

Each exposure is one NIRCam integration of 1800 s using 9 groups in DEEP8 (minimizes data rate while preserving cosmic-ray rejection). If a

parallel slot is longer, we scale exposure time linearly; if shorter, we may adopt MEDIUM readout to preserve 5 groups.

Tiering and filters.

Our tiered design matches the available mechanism moves per visit; filters are chosen to maximize dropout significance and suppress interlopers:

- Tier I (1 move; 60+ min):

Slot 1: SW F150W2 + LW F277W;

Slot 2: SW F150W2 + LW F444W.

F150W2 acts as an ultra-deep dropout (effectively stacking F115W+F150W+F200W); F277W/F444W provide the longest baseline redward of Ly.

- Tier II (2 moves): add F200W+F300M to refine the drop location (F200W) and flag strong-line boosters of F277W (F300M).
- Tier III (3 moves): add F210M+F410M to tighten photo-z and improve mid-z interloper control.
- Tier IV (≥ 4 moves): add F150W+F356W to further constrain SED shape and enable expanded legacy science.

Depth and selection thresholds.

Tier I requests 1800 s per configuration, reaching $F277W = 28.5$ (7 σ) and a 5 σ point-source limit of 6.5 nJy in F150W2, sufficient for selecting $mF277W < 27.8$ mag with $(F150W2 - F277W) > 2$ (3 σ) and $(F277W - F444W) < 1$.

Dithering.

We require at least three dithers per visit to ensure robust cosmic-ray rejection, accurate background subtraction, and uniform coverage. While we do not control the exact pattern in pure-parallel mode, most primary observations naturally satisfy 3 dithers.

Program scale (Cycle-4 analog).

Using the STScI Python package “jwpure” with realistic constraints (≥ 3 dithers, ≥ 1800 s/config, high |b|), we expect:

- Tier I (1 move): 46 visits 460 arcmin² (275 hr total)
- Tier II (2 moves): 31 visits 310 arcmin² (325 hr total)
- Tier III (3 moves): 8 visits 80 arcmin² (161 hr total)
- Tier IV (≥ 4 moves): 5 visits 150 arcmin² (216 hr total)

This distribution naturally delivers our ~ 0.3 deg² area while remaining fully compatible with primary programs that require longer filter sequences.

Proposal 10790 - Observation 1 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 1									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
Template	Observing Template: NIRCam Imaging									
	Module					Subarray				
Spectral Elements	ALL					FULL				
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F277W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 2 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 2 Diagnostic Status: No Diagnostics Observing Template: NIRCam Imaging										Wed May 27 20:00:14 GMT 2026
	Module					Subarray					
Template	ALL					FULL					
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F444W	MEDIUM2	6	1	1	1	558.312		

Proposal 10790 - Observation 3 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 3									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
Observing Template: NIRCam Imaging										
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F277W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 4 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 4									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
	Observing Template: NIRCam Imaging									
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F444W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 5 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 5									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
Observing Template: NIRCcam Imaging										
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F200W	F300M	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 6 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 6									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
Observing Template: NIRCcam Imaging										
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F277W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 7 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 7									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
	Observing Template: NIRCam Imaging									
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F444W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 8 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 8									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
Observing Template: NIRCam Imaging										
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F200W	F300M	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 9 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 9									
	Diagnostic Status: No Diagnostics									
Template	Observing Template: NIRCcam Imaging									
Spectral Elements	Module					Subarray				
	ALL					FULL				
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F210M	F410M	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 10 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 10									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
	Observing Template: NIRCam Imaging									
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F277W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 11 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 11 Diagnostic Status: No Diagnostics Observing Template: NIRCam Imaging Wed May 27 20:00:14 GMT 2026									
	Module					Subarray				
Template	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W2	F444W	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 12 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 12									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
Observing Template: NIRCam Imaging										
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F200W	F300M	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 13 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 13									Wed May 27 20:00:14 GMT 2026
	Diagnostic Status: No Diagnostics									
	Observing Template: NIRCam Imaging									
Template	Module					Subarray				
	ALL					FULL				
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F210M	F410M	MEDIUM2	6	1	1	1	558.312	

Proposal 10790 - Observation 14 - LAPIS: Large Area Parallel Imaging Survey

Observation	Proposal 10790, Observation 14									
	Diagnostic Status: No Diagnostics									
Template	Observing Template: NIRCam Imaging									
	Module					Subarray				
Spectral Elements	ALL					FULL				
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F356W	MEDIUM2	6	1	1	1	558.312	